

Marco Aurelio Costa

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EDUCATION



Chapman University

M.S. Computer Science & Electrical Engineering

Coursework: Statistical ML, Advanced AI, Bayesian Data Analysis, High Performance Computing, Database Systems

Orange, CA

Dec 2025

B.S. Computer Science, Minor: Information Security & Policy

Coursework: Algorithm Analysis, Database Management, Numerical Analysis, Linear Algebra, Statistics

May 2024

PROJECTS

Event Classification Tool for Sparse Video CV Pipelines

April 2026 - Present

TypeScript, React, Zod, C++, WebAssembly, Hyparquet

- Built a browser-based tool to define, debug, and label temporal events from sparse OCR and object-detection Parquet timelines
- Implemented manual classification and automated testing of classification configuration and metrics reporting
- Implemented Webassembly + C++ acceleration for a subset of operations for efficiency boosts when classifying large datasets
- Automated export feature selection, reducing total parameters by ~80% for more efficient downstream DL model training

Multi-Model Imitation Learning Pipeline for Behavior Cloning Balatro Agents

April 2026

Python, Linux, Bash, Jupyter, Ultralytics YOLOv8, PaddleOCR

- Built end-to-end imitation learning dataset on 500+ hours of downsampled video, generating 200M+ structured observations
- Trained object detection model achieving 0.995 mAP@0.5 on a synthetic dataset of 480K labeled instances across 12K images
- Trained character recognition model achieving 98.4% character accuracy on a synthetic dataset of 200K images
- Benchmarked OCR models and achieved 3.8× throughput improvement with <1% loss in character accuracy
- Ran distributed inference on machine learning cluster using 8 V100 GPUs to process 21M+ frames in under 60 hours
- Trained a behavioral cloning policy achieving 71% top-1 and 91% top-3 human-action prediction accuracy for live agent control

LLM-Based Data Curation & Analysis

August 2025

Python, Google Colab, OpenAI API

- Conducted market research by interviewing voluntary participants and converting interview transcripts into structured datasets
- Modified and used existing LLM analysis framework to conduct inductive thematic coding on large datasets of dialogue
- Extracted 800+ codes from 3 hours of dialogue and identified 3 themes for use in market analysis and product development

Analysis of Changing TV Show Norms Using Bayesian Modeling

December 2024

R, Python, BeautifulSoup, OpenAI API

- Used webscraping libraries to build dataset from 168 episode scripts with ~50K+ lines of dialogue
- Used GPT-4o to classify dialogue into structured content categories, creating labeled data for statistical modeling
- Applied linear regression to determine trends in content over ~20 years of airing
- Used MCMC to estimate uncertainty in model predictions of dialogue classifications
- Visualized trends and model outputs using ggplot2, enabling interpretable comparison between frequentist and Bayesian results

ACHIEVEMENTS

- Ranked Top 1,000 of 4 million global users in Hack The Box CTF competitions | [Profile](#)

September, 2021

SKILLS

- Programming: Python, C++, Java, SQL
- Machine Learning: PyTorch, Gymnasium, Imitation Learning, Gradient Boosting, Random Forest
- GPU / Parallel Computing: Multi-GPU Training, Distributed Inference, OpenMP, Open MPI
- Data & Systems: PySpark, Polars, Parquet, Multiprocessing, Data Pipelines
- Systems: AWS, Linux, Docker, Bash
- Computer Vision / OCR: OpenCV, YOLO, PaddleOCR, TrOCR, Object Tracking
- Languages: English (native), French (fluent), Italian (intermediate)

WORK EXPERIENCE

BioLumen - Business Operations & Data Analyst | San Francisco, CA

May 2024 - Present

- Estimated and visualized financial projections, identifying and comparing patterns across similar business operations
- Queried Shopify analytics data to identify anomalous customer purchase patterns and potential sources of fraud
- Debugged and maintained production web infrastructure across Bluehost, Squarespace, and Google Workspace